1 $3(4) - 2(-5)$ or $12 - -10$ or $12 + 10$ **[M1]**

22 **[A1]**

[Total Marks: 2]

2 Substitute the given values into the formula: $v = 5 + (-2)(3)$ **[M1]**

$v = 5 - 6$ **[M1]**

-1 **[A1]**

[Total Marks: 3]

3 (a) $4(-2) - 3(5)$ or $-8 - 15$ **[M1]**

-23 **[A1]**

(b) $\frac{5^2 - 7}{2(-3)}$ or $\frac{25 - 7}{-6}$ **[M1]**

-3 **[A1]**

[Total Marks: 4]

4 (a) Substitutes values correctly: $5(3) - 2(-4)$ **[M1]**

$15 + 8 = 23$ **[A1]**

(b) Substitutes values: $\frac{(-4)^2 - (2)^3}{2(3)}$ **[M1]**

Evaluates numerator: $16 - 8 = 8$

$\frac{8}{6} = \frac{4}{3}$ or $1\frac{1}{3}$ **[A1]**

[Total Marks: 4]

5 (a) (a) $40 + 25(3.5)$ or $40 + 87.5$ **[M1]**

127.50 **[A1]**

(b) $-12 + 9.8 \times 3$ **[M1]**

$-12 + 29.4$ **[M1]**

17.4 **[A1]**

[Total Marks: 5]

6 (a) $d = \sqrt{(-8 - 4)^2 + (2 - -3)^2}$ or $\sqrt{144 + 25}$ **[M1]**

13 **[A1]**

(b) $17 = \sqrt{(3 - k)^2 + (-7 - 8)^2}$ **[M1]**

$289 = (3 - k)^2 + 225$ **[M1]**

$(3-k)^2 = 64$ or $k^2 - 6k - 55 = 0$ **[M1]**

$k = -5$ and $k = 11$ **[A1]**

[Total Marks: 6]

7 **(a)** States $A =$ Curved Surface Area + Base Area $= 2\pi rh + \pi r^2$ **[M1]**

Factorises πr out to obtain $\pi r(2h + r)$ **[A1]**

(b) **(i)** Substitutes $r = 4$ and $h = 6$: $A = \pi(4)(2(6) + 4)$ **[M1]**

$A = 4\pi(12 + 4) = 64\pi$ **[A1]**

(ii) Rearranges volume formula: $h = \dfrac{V}{\pi r^2}$ **[M1]**

Substitutes h into area formula: $A = 2\pi r\left(\dfrac{V}{\pi r^2}\right) + \pi r^2$ **[M1]**

Simplifies first term: $\dfrac{2\pi rV}{\pi r^2} = \dfrac{2V}{r}$, resulting in $A = \dfrac{2V}{r} + \pi r^2$ **[A1]**

[Total Marks: 7]

8 **(a)** Substitute $x = -2$ and $y = 4$: $P = \dfrac{2(-2)^2 - 3(4)}{-2 - 4}$ **[M1]**

$P = \dfrac{2(4) - 12}{-6} = \dfrac{8 - 12}{-6}$ **[M1]**

$\dfrac{2}{3}$ (or equivalent) **[A1]**

(b) Multiply both sides by $(x - y)$: $P(x - y) = 2x^2 - 3y$ **[M1]**

Expand: $Px - Py = 2x^2 - 3y$ **[M1]**

Collect terms with y on one side: $3y - Py = 2x^2 - Px$ **[M1]**

Factorise and divide: $y(3 - P) = 2x^2 - Px \implies y = \dfrac{2x^2 - Px}{3 - P}$ (or equivalent) **[A1]**

[Total Marks: 7]

9 **(a)** $V = \dfrac{1}{3}\pi(6.4)^2(12.6)$ **[M1]**

540.5 **[A1]**

(b) $800 = \dfrac{1}{3}\pi r^2(14)$ **[M1]**

$r^2 = \dfrac{2400}{14\pi}$ or $54.567\ldots$ **[M1]**

7.39 **[A1]**

[Total Marks: 5]

10 **(a)** Substitute $h = 3.5$: $C = 45 + 35(3.5)$ **[M1]**

167.5 **[A1]**

(b) Substitute $C = 325$: $325 = 45 + 35h$ **[M1]**

$280 = 35h$ **[M1]**

8 **[A1]**

[Total Marks: 5]

11 (a) Substitutes $h = 4.2$ and $a = 6.5$ into formula: $V = \frac{1}{6}\pi(4.2)(3(6.5)^2 + 4.2^2)$ **[M1]**

$V = 317.53... = 317.5$ **[A1]**

(b) Multiplies by 6 and divides by πh: $\frac{6V}{\pi h} = 3a^2 + h^2$ **[M1]**

Subtracts h^2: $\frac{6V}{\pi h} - h^2 = 3a^2$ **[M1]**

$a^2 = \frac{2V}{\pi h} - \frac{h^2}{3}$ or $a^2 = \frac{1}{3}\left(\frac{6V}{\pi h} - h^2\right)$ **[A1]**

(c) $a = \sqrt{\frac{2(150)}{\pi(3)} - \frac{3^2}{3}} = \sqrt{31.83... - 3} = \sqrt{28.83...} = 5.37$ **[B1]**

[Total Marks: 6]

12 (a) (a) $5.2(4.5) + 0.5(-9.8)(4.5)^2$ **[M1]**

-75.825 **[A1]**

(b) $-9.8 = 4.9t + \frac{1}{2}(-9.8)t^2$ **[M1]**

$4.9t^2 - 4.9t - 9.8 = 0$ or $t^2 - t - 2 = 0$ **[M1]**

$t = 2$ (rejects -1 as time must be positive) **[A1]**

[Total Marks: 5]

13 Substitution of all three values correctly e.g. $\frac{3(5) - (-4)^2}{2(0.5)}$ **[M1]**

-1 **[A1]**

[Total Marks: 2]

14 (a) $85(7) + 0.24(450)$ **[M1]**

703 **[A1]**

(b) $1475.60 = 85(14) + 0.24m$ **[M1]**

$1475.60 - 1190 = 0.24m$ (or $285.60 = 0.24m$) **[M1]**

1190 **[A1]**

[Total Marks: 5]

15 **(a)** **(i)** $\dfrac{6^2-(-2)^2}{6(-2)}$ **[M1]**

$-\dfrac{8}{3}$ **[A1]**

(ii) $\dfrac{0.5^2-4^2}{0.5(4)}$ **[M1]**

$\dfrac{-15.75}{2}$ **[M1]**

$-\dfrac{63}{8}$ **[A1]**

(b) $\dfrac{x^2-2^2}{2x}=\dfrac{3}{2}$ **[M1]**

$2(x^2-4)=6x$ (or $x^2-3x-4=0$) **[M1]**

4 **[A1]**

[Total Marks: 8]

16 Correct substitution: $5(-2)^2-3(4)$ or $5(4)-12$ **[M1]**

8 **[A1]**

[Total Marks: 2]

17 **(a)** $45+32h$ (or equivalent) **[B1]**

(b) 157 **[B1]**

(c) Correct substitution: $(-3)^3-4(-3)(2.5)$ or $-27-(-30)$ **[M1]**

3 **[A1]**

[Total Marks: 4]

18 Numerator correctly evaluated: $-4.5-(-3.2)^2=-4.5-10.24=-14.74$ **[B1]**

Denominator correctly evaluated: $1.8-\sqrt{0.64}=1.8-0.8=1$ **[B1]**

-14.74 **[B1]**

[Total Marks: 3]

19 **(a)** -6 or -20 seen **[M1]**

-26 **[A1]**

(b) $(-3)^2=9$ or $4(-3)=-12$ seen **[M1]**

-3 **[A1]**

[Total Marks: 4]

20 $\pi \times (4.5)^2 \times 8.2 + \frac{2}{3} \times \pi \times (4.5)^3$ **[M1]**

$521.6... + 190.8...$ or $712.5...$ **[M1]**

713 **[A1]**

[Total Marks: 3]

21 **(a)** $40 + 25h$ **[B1]**

(b) $(-2)^3 = -8$ and $(-3)^2 = 9$ **[M1]**

$\sqrt{\frac{1}{16}} = \frac{1}{4}$ **[B1]**

-68 **[A1]**

[Total Marks: 4]